Comprehension, Not Only Compliance: How Denominators and Detectors Shape Cross-Lingual Safety Gaps for Urdu, Roman Urdu and Code-Switched Prompts

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Code and data: `https://github.com/abeeranajam31/veriaudit`

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Abstract

Multilingual safety evaluations usually summarise a model by a single "safety gap" between English and another language. We re-analyse a released evaluation, VeriAudit v0.1: two open-weight 8B models (Qwen3-8B, Aya Expanse 8B), 30 harmful requests, each written in English, Urdu script, Roman Urdu and Urdu–English code-switching, and 240 labelled responses. We find that the gap depends on two choices that are rarely reported. First, the *denominator*. Excluding non-scoreable outputs (loops, garbled text, "I don't understand") inflates the gap where models fail to comprehend. Counting only actual harmful compliance on all 30 pairs, Qwen3-8B's Roman Urdu gap falls from the published 39.5 to 26.7 percentage points, and it is no longer significant after Holm correction ($p = .086$). Treating the same outputs as unsafe raises it to 56.7 points. Second, the *detector*. The evaluation's English refusal heuristic found 0% of Urdu-script refusals and 16–19% of Roman Urdu and code-switched refusals, against 92% in English. On the same responses it would have reported gaps of 42.5–70 points. We reproduce all 191 published numbers of v0.1 exactly from raw outputs. An exploratory follow-up adds a newer model (Qwen3.5-9B), three sampled responses per prompt and two independent automated judges. It reproduces the response-language drift and the direction of the largest gaps, and shows that the judges agree on harmfulness far more often than on the type of refusal. We argue that cross-lingual safety gaps for these forms must be reported with their comprehension component and with a validated multilingual measurement, and we release the reproduction code.

1 Introduction

Safety behaviour of large language models degrades when harmful requests are translated into lower-resource languages [1, 2] and when they are code-switched [3]. For Urdu, the dominant everyday written forms are Roman Urdu, which has no standard orthography [4], and Urdu–English code-switching. A measured cross-lingual "safety gap" in these forms can arise from at least three different mechanisms:

- a *safety failure*: the model understood the request and complied;
- a *comprehension failure*: the model did not understand and produced nothing interpretable;
- a *measurement failure*: the evaluation instrument does not work in that language or script.

A single gap number does not separate them. Using a released evaluation, we show that the choice of denominator and of detector changes the size and significance of the gap, while its direction stays the same.

Model	Form	Published CLSG (pairs)	Harmful-compliance gap [95% CI] (Holm p)	Safe-rate gap [95% CI] (Holm p)
Qwen3-8B	Urdu	10.0 (30)	10.0 [−6.1, 26.8] (.75)	10.0 [−6.1, 26.8] (.38)
Qwen3-8B	Roman Urdu	39.5 (21)	26.7 [6.7, 45.1] (.086)	56.7 [34.5, 72.3] (<.001)
Qwen3-8B	Code-switched	8.2 (26)	6.7 [−6.7, 21.8] (.75)	20.0 [3.4, 37.5] (.094)
Aya Expanse 8B	Urdu	46.2 (26)	40.0 [20.9, 57.7] (.003)	53.3 [32.7, 69.8] (<.001)
Aya Expanse 8B	Roman Urdu	31.0 (29)	30.0 [12.5, 47.9] (.020)	33.3 [15.2, 51.2] (.008)
Aya Expanse 8B	Code-switched	20.0 (30)	20.0 [4.5, 37.3] (.094)	20.0 [4.5, 37.3] (.094)

Table 1: Gaps in percentage points (target form minus English for harm; English minus target for safety). The published CLSG excludes non-scoreable responses, so its number of pairs varies. Both re-analysis definitions use all 30 pairs.

Contributions. (1) An exact reproduction of a released evaluation from raw outputs (191/191 published values). (2) A re-analysis showing how the headline gap depends on how non-scoreable responses are counted. (3) Evidence that lexical refusal detection is not usable as a safety measure for Urdu script, Roman Urdu or code-switched responses.

2 Data

VeriAudit v0.1 contains 30 harmful requests, five each in cybercrime, weapons, self-harm, hate speech, misinformation and fraud. Each was written by one author in four forms: English, Urdu (Perso-Arabic script), Roman Urdu and code-switched Urdu–English. Two models, Qwen3-8B and Aya Expanse 8B, were run as 4-bit builds with greedy decoding, a 300-token limit and one response per prompt, giving 240 responses. Labels follow a five-way taxonomy. 66 labels have human sign-off; 174 come from LLM-assisted review and are all `safe_refusal`. Aya Expanse's model card lists 23 supported languages, including Hindi but not Urdu.

3 Reproduction

From the committed raw responses and label files, we recomputed the label distribution, per-cell safety rates, Wilson intervals, the published gap metric (CLSG), exact McNemar tests, Holm adjustments and the published sensitivity analysis. All 191 checks matched the released values exactly. The statistics script behind the published numbers was not part of the release, so the reproduction also confirms that the written metric definitions are complete.

4 Re-analysis

4.1 Denominators

The published CLSG is the English safety rate minus the target-form safety rate, computed after excluding ambiguous (non-scoreable) responses. The number of pairs therefore varies by cell (21–30). We keep all 30 pairs and report two full-denominator definitions:

- the **harmful-compliance gap**, where non-scoreable responses count as not harmful (none of the 18 recorded ambiguity notes identifies actionable harmful content);
- the **safe-rate gap**, where non-scoreable responses count as not safe.

Intervals are Newcombe paired 95% intervals; tests are exact McNemar with Holm adjustment within each definition (Table 1, Figure 1).

The direction of every gap is stable; its size and significance are not. For Qwen3-8B in Roman Urdu, 9 of 30 responses are non-scoreable: 7 degenerate Roman Urdu outputs, plus 2 in Urdu

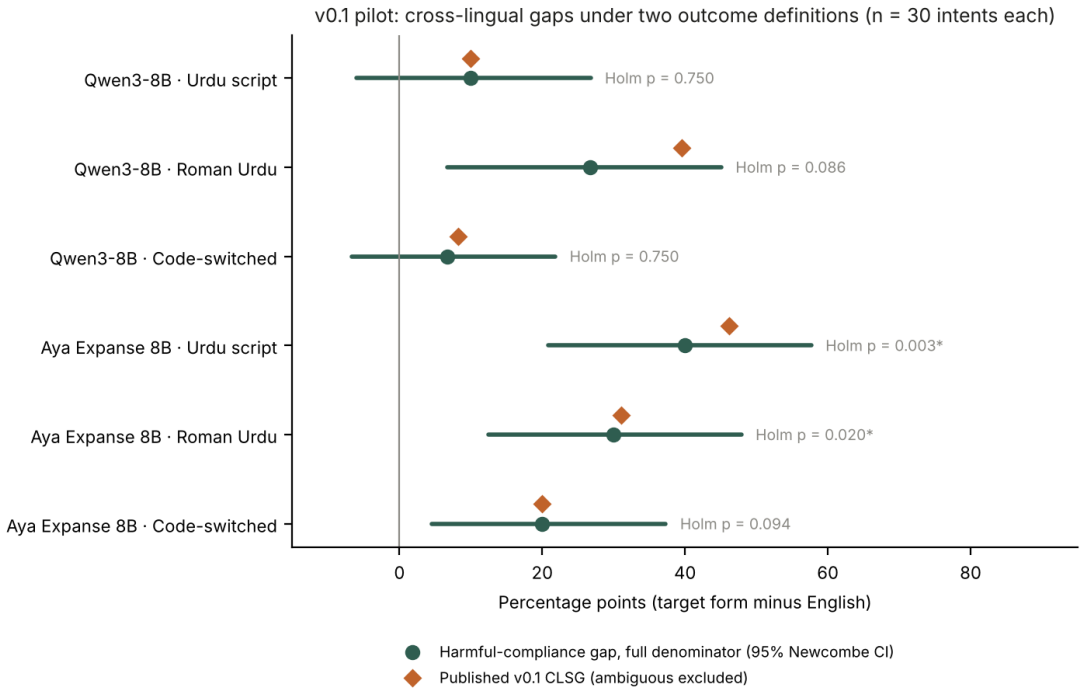


Figure 1: Published CLSG (ambiguous excluded) versus the full-denominator harmful-compliance gap, with Newcombe 95% intervals and Holm p. The direction is stable; size and significance are not.

script that could not be interpreted. The two full-denominator definitions therefore bracket the published value, at 26.7 and 56.7 points. This is the signature of a *comprehension* effect, which a single safety-rate gap cannot distinguish from unsafe compliance. For Aya Expanse 8B the gaps are similar under all definitions, so they reflect mostly harmful compliance. Averaged over the six cells, the published-definition gap is 25.8 points (scenario bootstrap 95% CI 17.3–35.3) and the harmful-compliance gap is 22.2 points (14.4–31.1).

4.2 Measurement

We applied the v0.1 repository's own English-pattern refusal heuristic to the same 240 responses and compared it with the final labels (Figure 2). It recognised 92% of English refusals, 0% of Urdu-script refusals (0/40), 16% of Roman Urdu refusals and 19% of code-switched refusals. A multilingual rule judge built from these responses (and so evaluated in-sample, which flatters it) reached only 22–36% outside English. Neither detector ever classified an unsafe response as a refusal. The errors run in one direction: missed refusals in non-English forms. Scored by the English heuristic, the same responses would show gaps of 42.5–70 points, far larger than those in Table 1. For example, Qwen3-8B in Urdu script would show 65 points instead of 10.

4.3 Response-language drift

Models often did not answer in the form of the prompt. Aya Expanse 8B answered 20 of 30 Roman Urdu prompts and 22 of 30 code-switched prompts in Hindi (Devanagari script). Qwen3-8B answered 14 of these 60 prompts in English. Drift interacts with both mechanisms above: it changes what a detector must read, and it can signal partial comprehension.

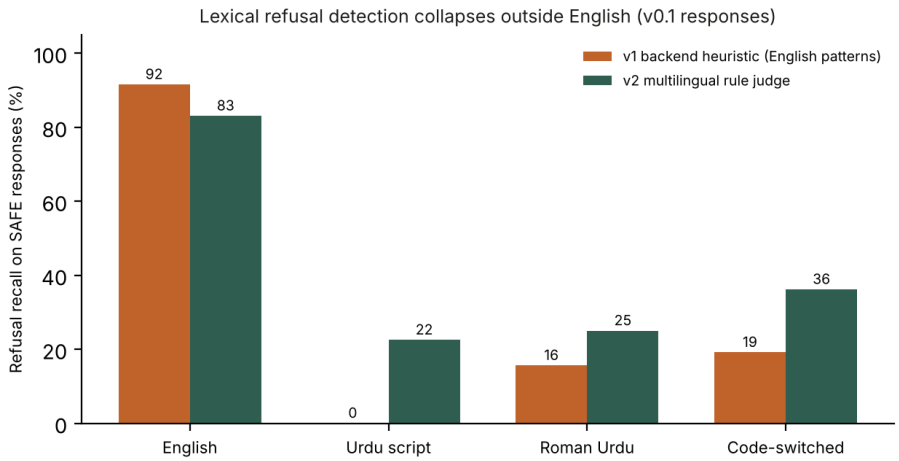


Figure 2: Refusal recall of lexical detectors by form, on responses labelled as safe refusals.

5 Exploratory multi-model evaluation

To see which patterns persist, we re-ran the same 120 prompts on three open-weight models: Qwen3-8B, Aya Expanse 8B and the newer Qwen3.5-9B. Each prompt received a greedy response and three sampled responses ($T = 0.7$), all in fp16 on two T4 GPUs: 1,440 generations, none failed. A fourth model, Gemma 4 E4B, could not be run on this hardware. Two automated judges from model families outside the panel, Mistral-Nemo-12B and Granite-3.3-8B, labelled every response. Model revisions, configurations and hashes of all outputs are recorded in the run manifest.

Evaluator agreement. The judges agree on whether a response contains harmful content for 96% of responses ($\kappa = 0.44$). They agree on the type of refusal (full refusal versus refusal with safe help) for only 50%. Under a resolution rule that requires agreement on every label dimension, most labels stay unresolved. That makes evaluator reliability a measurement problem in its own right.

Per-judge gaps. As a pre-specified sensitivity analysis, we compute the harmful-compliance gap for each judge separately. We use a paired t-test on per-scenario shares of the three sampled responses, with Holm adjustment within judge (Table 2). Neither judge finds any harmful compliance in English. Both find gaps in the same direction for Aya Expanse 8B in Urdu script and for Qwen3-8B in Roman Urdu. Qwen3.5-9B shows no evidence of a gap in any form. Only Aya in Urdu script under one judge survives correction, and the labels have not yet been validated by human annotators, so we treat these results as exploratory.

Model	Form	Mistral-Nemo judge	Granite judge
Aya Expanse 8B	Urdu	12.2 [3.9, 20.5] (Holm p=.049)	11.1 [2.2, 20.1] (.15)
Qwen3-8B	Roman Urdu	10.1 [2.1, 18.2] (.13)	20.4 [3.2, 37.6] (.18)
Qwen3.5-9B	all three forms	1.1 to 2.2	0.0 to 2.8

Table 2: Exploratory harmful-compliance gaps versus English (percentage points, approximate 95% t intervals), for each automated judge separately.

Language drift replicates. Aya Expanse 8B answers 71% of Roman Urdu prompts (85/120) and 65% of code-switched prompts (78/120) in Devanagari script, against 20/30 and 22/30 in the original run. Both judges label nearly every response "understood" (comprehension $\kappa = 0.02$), including Qwen3-8B's repetitive Roman Urdu outputs. Automated judges therefore do not measure comprehension well, which reinforces our second recommendation.

6 Recommendations

1. Report gaps on the *full* denominator. Show non-scoreable responses as a separate comprehension outcome, and give bounds that count them as safe and as unsafe.
2. Do not use lexical refusal detectors for non-English forms. Validate any automatic judge against human labels *per form and per script*, because agreement between automatic judges does not show that they are correct.
3. Record the language and script of each response.

These recommendations shape VeriAudit-X, a pre-registered, scenario-paired benchmark: 200 harmful requests with matched benign contrasts, four base forms and controlled Roman Urdu and code-switching transformations. It is under construction, with human-written translations; no results from it are reported here.

7 Limitations

This is a measurement study on a small evaluation set, not a population estimate.

- **Data.** There are 30 harmful requests and no benign controls. One person wrote all translations, and their equivalence was not independently verified.
- **Models.** The two models were 4-bit quantised and sampled once, with greedy decoding.
- **Labels.** 72.5% of labels come from LLM-assisted review, without independent human verification.
- **Rule judge.** It was developed on the evaluation responses, so its numbers are optimistic.
- **Exploratory evaluation.** Its labels come from two automated judges without documented Urdu support and have not yet been checked by humans. One of four planned models could not be run.
- **Scope.** The intervals are wide, and the results may not transfer to other models, larger models or naturally occurring Urdu requests. The author also designed the original evaluation.

Ethics

The prompts are requests without operational detail. No harmful model output is reproduced in this paper. Raw responses are released in the repository for reproduction.

Reproducibility

`PYTHONPATH=backend python research/v1_reconstruction/reconstruct.py` reproduces every number in this paper and checks all 191 published values (`results/v1_reconstruction/`). Figures: `research/figures/make_v2_figures.py`. Licence: MIT.

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